

1. Let R be the focus of the parabola $y^2 = 20x$ and the line $y = mx + c$ intersect the parabola at two points P and Q . Let the points $G(10, 10)$ be the centroid of the triangle PQR . If $c - m = 6$, then PQ^2 is

[2023 (08 Apr Shift 1)]

- (1) 296
(2) 325
(3) 317
(4) 346

2. The ordinates of the points P and Q on the parabola with focus $(3, 0)$ and directrix $x = -3$ are in the ratio

3 : 1. If $R(\alpha, \beta)$ is the point of intersection of the tangents to the parabola at P and Q , then $\frac{\beta^2}{\alpha}$ is equal to

[2023 (08 Apr Shift 2)]

3. Let a common tangent to the curves $y^2 = 4x$ and $(x - 4)^2 + y^2 = 16$ touch the curves at the points P and Q . Then $(PQ)^2$ is equal to _____.

[2023 (10 Apr Shift 1)]

4. Let PQ be a focal chord of the parabola $y^2 = 36x$ of length 100, making an acute angle with the positive x -axis. Let the ordinate of P be positive and M be the point on the line segment PQ such that $PM : MQ = 3 : 1$. Then which of the following points does NOT lie on the line passing through M and perpendicular to the line PQ ?

[2023 (13 Apr Shift 1)]

- (1) $(-6, 45)$
(2) $(6, 29)$
(3) $(3, 33)$
(4) $(-3, 43)$

ANSWER KEYS

1. (2)

2. (16)

3. (32)

4. (4)

1. (2)

Given,

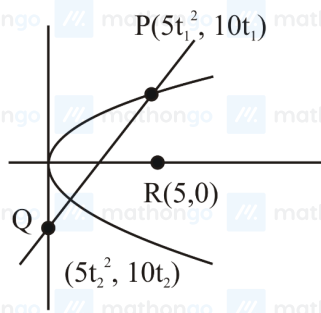
R be the focus of the parabola $y^2 = 20x$ and the line $y = mx + c$ intersect the parabola at two points P and Q ,

And the points $G(10, 10)$ be the centroid of the triangle PQR ,

Now focus of the parabola $y^2 = 20x$ will be, $R(5, 0)$

And parametric points of PQ be $(5t^2, 10t)$,

Now plotting the diagram we get,



Now finding the centroid of the triangle we get,

For x -coordinate we get,

$$\Rightarrow \frac{5t_1^2 + 5t_2^2 + 5}{3} = 10$$

$$\Rightarrow t_1^2 + t_2^2 = 5 \quad \dots (i)$$

Now for y -coordinate we get,

$$\frac{10(t_1 + t_2)}{3} = 10$$

$$\Rightarrow t_1 + t_2 = 3 \quad \dots (ii)$$

Now solving both equations we get, $t_1 = 1, t_2 = 2$

So, points will be, $P \equiv (5, 10)$ & $Q \equiv (20, 20)$

Hence, equation of $PQ = y - 10 = \frac{10}{15}(x - 5)$

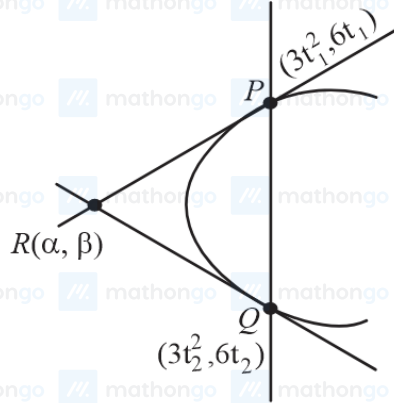
$$\Rightarrow 3y - 30 = 2x - 10$$

$$\Rightarrow y = \frac{2}{3}x + \frac{20}{3}, \text{ so on comparing with } y = mx + c, \text{ we get } c - m = 6$$

Hence, $PQ^2 = 225 + 100 = 325$

2. (16)

Given,

Focus $(3, 0)$, Directrix, $x = -3$ So, the equation of parabola will be, $y^2 = 12x$ Now taking the parametric point $(3t^2, 6t)$ and plotting the diagram we get,Now we know that $R(\alpha, \beta)$ will be the arithmetic mean of ordinate and geometric mean of abscissa

will be,

$$\text{Hence, } \beta = \frac{6(t_1 + t_2)}{2} \text{ and } \alpha = \sqrt{3t_1^2 \times 3t_2^2} = 3t_1t_2$$

$$\text{So, putting the value in } \frac{\beta^2}{\alpha} \Rightarrow \frac{\beta^2}{\alpha} = \frac{9(t_1 + t_2)^2}{3t_1t_2} = \frac{9(4t_2)^2}{3 \times 3t_2^2} = 16 \left(\because \frac{6t_1}{6t_2} = 3, \text{ given} \right)$$

3. (32)

Given,

Common tangent to the curves $y^2 = 4x$ and $(x - 4)^2 + y^2 = 16$ touch the curves at the points P and Q

Let, the equation of tangent be,

$$y = mx + \frac{a}{m}, \text{ where } a = 1 \text{ is focus and } m \text{ is slope,}$$

$$\text{So, equation of tangent to parabola will be, } y = mx + \frac{1}{m}$$

Also line is tangent to circle,

So, using perpendicular distance formula from centre $(4, 0)$ to tangent which will be radius $r = 4$, we get,

$$\left| \frac{4m + \frac{1}{m}}{m^2 + 1} \right| = 4$$

$$\Rightarrow 16m^2 + \frac{1}{m^2} + 8 = 16m^2 + 16$$

$$\Rightarrow m^2 = \frac{1}{8}$$

$$\Rightarrow m = \pm \frac{1}{2\sqrt{2}}$$

$$\text{Now if } m = \frac{1}{2\sqrt{2}} \text{ then } P = \left(\frac{a}{m^2}, \frac{2a}{m} \right) = (8, 4\sqrt{2})$$

$$\text{Now equation of tangent will be, } x - 2\sqrt{2}y + 8 = 0$$

$$\text{And } Q \text{ will be foot of perpendicular from } (4, 0) \text{ on tangent } x - 2\sqrt{2}y + 8 = 0,$$

So, using foot of perpendicular formula we get,

$$\frac{x-4}{1} = \frac{y-0}{-2\sqrt{2}} = -\frac{4-0+8}{1^2 + (2\sqrt{2})^2}$$

$$\Rightarrow (x, y) = \left(\frac{8}{3}, \frac{8\sqrt{2}}{3} \right)$$

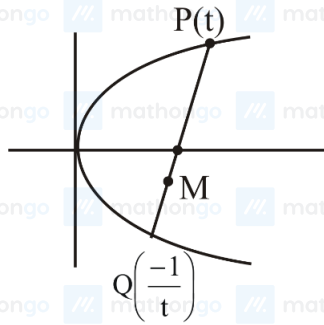
Hence, by distance formula we get,

$$(PQ)^2 = \left(\frac{16}{3} \right)^2 + \left(\frac{4\sqrt{2}}{3} \right)^2$$

$$\Rightarrow (PQ)^2 = \frac{256+32}{9} = 32$$

4. (4)

We have been given the length of focal chord $PQ = 100$.



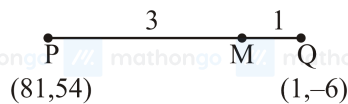
Length of focal chord at $(t) = a\left(t + \frac{1}{t}\right)^2 = 100$

Where $a = 9$

$$t + \frac{1}{t} = \pm \frac{10}{3}$$

$$\therefore t = 3, \frac{1}{3}, -3, -\frac{1}{3}$$

Since ordinate of P is +ve



$$\therefore t = 3 \text{ or } \frac{1}{3}$$

($t = 3$ as slope of PQ is positive)

$$\therefore M(21, 9)$$

$$\text{Required line: } y - 9 = \frac{-80}{60}(x - 21)$$

$$\Rightarrow 4x + 3y = 111$$

$(-3, 43)$ does not lie on line.

Hence this is the correct option.